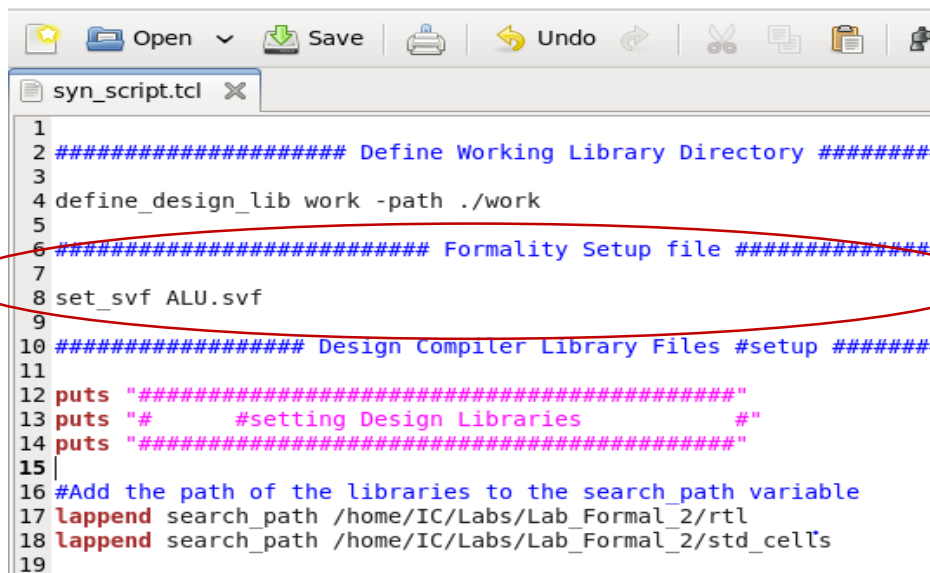


Lab_Formal_2

You have to check the design equivalence of the generated netlist post synthesis Verses RTL Verilog files using Formal Verification Techniques.

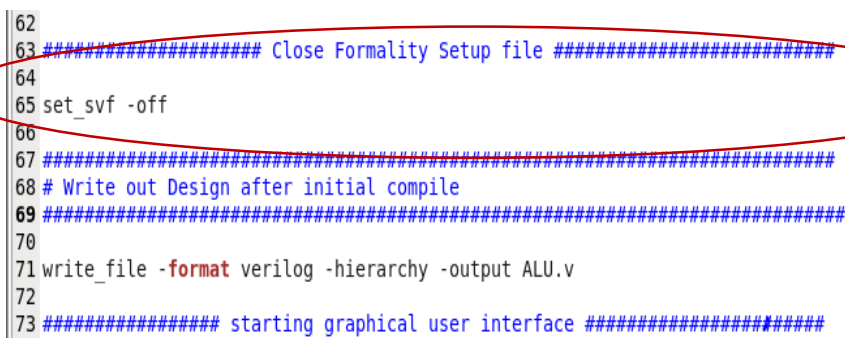
Steps: -

1. Copy **Lab_Formal_2** folder from your computer into the virtual machine inside **Labs** directory
2. Open **syn_script.tcl**
 1. Add the following command under section “**Formality Setup File**” to generate svf file used in formal verification
 - i. **set_svf ALU.svf**



```
1
2 ##### Define Working Library Directory #####
3
4 define_design_lib work -path ./work
5
6 ##### Formality Setup file #####
7
8 set_svf ALU.svf
9
10 ##### Design Compiler Library Files #setup #####
11
12 puts "#####"
13 puts "#      #setting Design Libraries      #"
14 puts "#####"
15
16 #Add the path of the libraries to the search_path variable
17 lappend search_path /home/IC/Labs/Lab_Formal_2/rtl
18 lappend search_path /home/IC/Labs/Lab_Formal_2/std_cells
19
```

2. Add the following command before write command under section “**Close Formality Setup File**” to stop writing in svf file and close it



```
62
63 ##### Close Formality Setup file #####
64
65 set_svf -off
66
67 #####
68 # Write out Design after initial compile
69 #####
70
71 write_file -format verilog -hierarchy -output ALU.v
72
73 ##### starting graphical user interface #####
74
```

2. Run synthesis using the following command to generate formality setup file “**SVF File**”
 - **dc_shell -f syn_script.tcl | tee syn.log**

3. Open fm_script.tcl and add the following commands: -

1) Formality Guidance

- set synopsys_auto_setup true
- set_svf design.svf

2) Load Reference Design

- Load technology libraries in db format
- Read RTL Files
- Set Top Design

3) Load Implementation Design

- Load technology libraries in db format
- Read netlist post synthesis
- Set Top Design

4) match

5) verify

6) generate reports

7) Run the script in fm folder using the following command

- **fm_shell -f fm_script.tcl | tee fm.log**

8) Check that all the compare points are succeeded and there are no failing points